

Ajit Sharma

Backend Engineer | Python | FastAPI | REST APIs | Distributed Systems

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Professional Summary

Backend engineer with experience building API-driven systems and scalable data processing pipelines using Python. Experienced in designing REST APIs, implementing asynchronous workflows, and developing backend services that integrate with external systems. Developed a FastAPI-based email validation API to explore production-style backend architecture including request validation, authentication, rate limiting, and external service orchestration. Strong experience with Python concurrency and parallel data processing.

Technical Skills

- Programming: Python
- Backend Development: FastAPI, Pydantic, REST API design, External API integrations, Authentication, Rate limiting
- Concurrency: Python multiprocessing, Multithreading, Parallel data processing
- Databases: PostgreSQL
- Development Workflow: Git, Code reviews, AI-assisted development tools
- Observability: Structured logging, Health checks, Debugging using logs

Backend Engineering Project

Email Validation API (FastAPI) — Backend Engineering Project | 2024 – Present

- Designed REST API endpoints using FastAPI with request and response validation using Pydantic.
- Implemented asynchronous workflows integrating external email verification providers.
- Designed internal service logic to orchestrate multiple verification providers and aggregate responses.
- Implemented API key authentication and rate limiting for secure access and service stability.
- Added timeout handling and retry strategies for unreliable external dependencies.
- Implemented structured logging to trace request lifecycle and support debugging.
- Used AI-assisted development tools to accelerate development while maintaining architectural ownership.

Professional Experience

Software Engineer — SAP | Apr 2022 – Present

- Contributed to backend components within enterprise software systems.
- Implemented backend features within structured development workflows.
- Investigated and resolved production issues using logs and debugging.
- Collaborated with engineering teams to maintain system reliability.

Scholar Engineer (Work + M.Tech Program) — SAP | Aug 2020 – Apr 2022

- Scaled a computer vision system detecting mask and social distancing violations using CCTV footage from 5 to 36 cameras.
- Implemented Python multiprocessing and multithreading to parallelize image processing workloads.
- Worked on an image processing pipeline detecting surface defects in industrial metal components.
- Developed a Microsoft Outlook add-in integrating email workflows with SAP Business ByDesign ERP.

Education

M.Tech – Software Engineering, BITS Pilani